

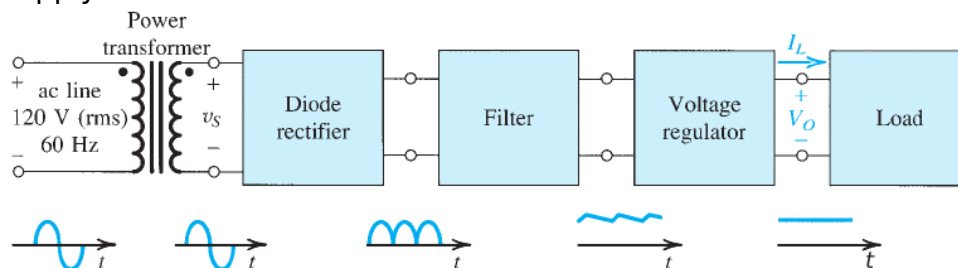
Lecture 3a

EE-215 Electronic Devices and Circuits

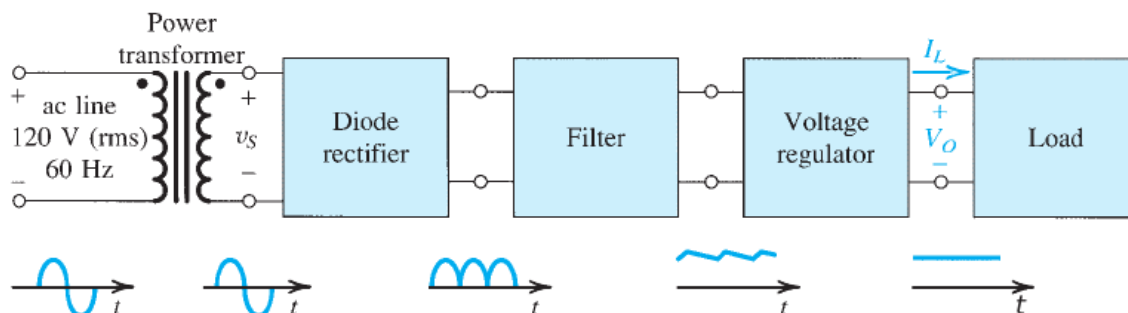
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Rectifier Circuits

- An important application of diodes is in the design of rectifier circuits
 - A diode rectifier forms an essential building block of a dc power supply



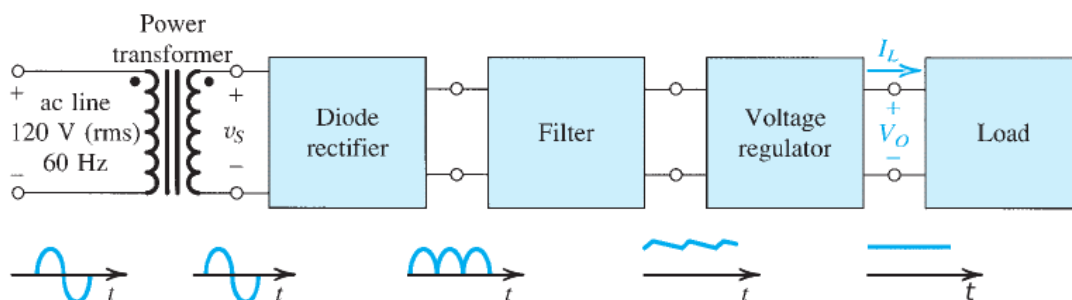
- **Figure 4.20** Block diagram of a dc power supply.
- A dc power supply produce a dc voltage,
 - which is required to power essentially every electronic device,
 - including personal computers, televisions and stereo systems
- Also battery chargers for portable electronic devices such as cellphones, laptop computers etc. contain rectifiers.
- The dc power supply is fed from ac line typically (230V rms 50Hz) or (120V rms 60Hz)
 - and it delivers a dc voltage V_O to an electronic circuit represented here by the load block.
 - The building blocks of a dc power supply are
 - power transformer
 - diode rectifier
 - filtering stage
 - voltage regulator



- **Figure 4.20** Block diagram of a dc power supply.
- The first block of a dc power supply is the power transformer
 - it is used to step down (or step up) the line ac voltage (of 120V or

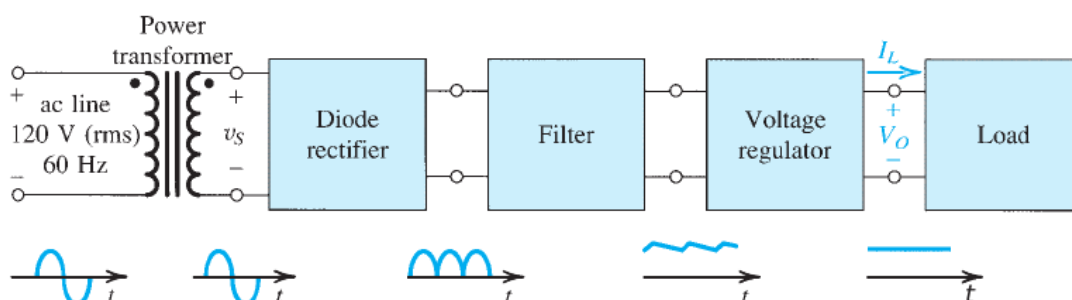
230V rms)

- to the desired ac voltage required to yield the particular dc voltage output of the supply
- for example, a secondary voltage of 8V rms may be required for a dc output of 5V,
- which can be achieved by using a transformer with turn ratio of 15:1 (if line voltage is 120V rms)
- The power transformer also provide
 - electrical isolation between the electronic equipment and the ac line.
 - this isolation reduces the risk of electric shock to the equipment user.



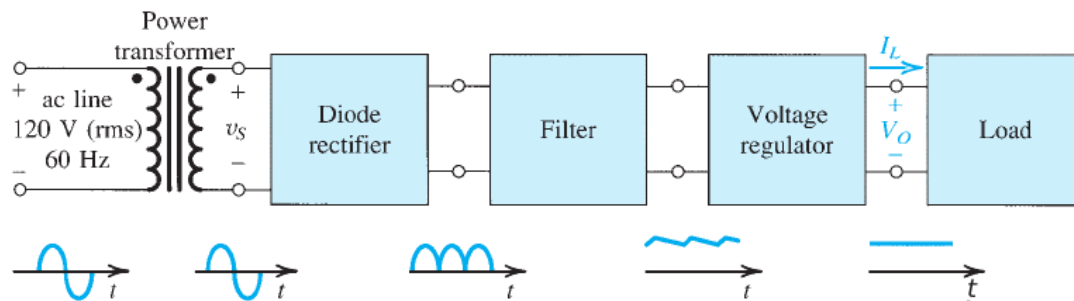
● **Figure 4.20** Block diagram of a dc power supply.

- The diode rectifier then
 - convert the ac voltage v_S to a unipolar output voltage
- this waveform has a non-zero average component,
 - but it is pulsating
 - and can not be readily used as a dc source for electronic circuits.



● **Figure 4.20** Block diagram of a dc power supply.

- Thus a filter block is required
 - to minimize the variations in the
 - magnitude of the rectifier output.
- A single capacitor can be used as an output filter.
- The output of the filter block is thus much more constant
 - than without the filter, but it still has a
 - time-dependent component called ripple.

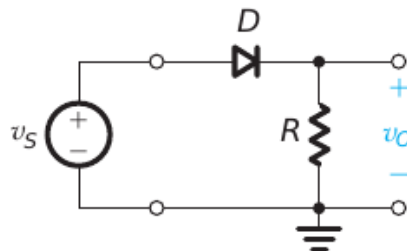


• **Figure 4.20** Block diagram of a dc power supply.

- This time-dependent component, “ripple ” is minimized by using a voltage regulator.
- The voltage regulator not only minimize the ripple component but also provide load regulation ($\frac{\Delta v_O}{\Delta I_L}$)

The Half-Wave Rectifier

- utilizes alternate half-cycles of the input sinusoid



- (a)
- using the constant-voltage-drop model for the diode
- $\Rightarrow v_O = \begin{cases} 0 & , v_S < V_D \\ v_S - V_D & , v_S \geq V_D \end{cases}$
- where V_D is the constant voltage drop across the conducting diode, typically around 0.7V

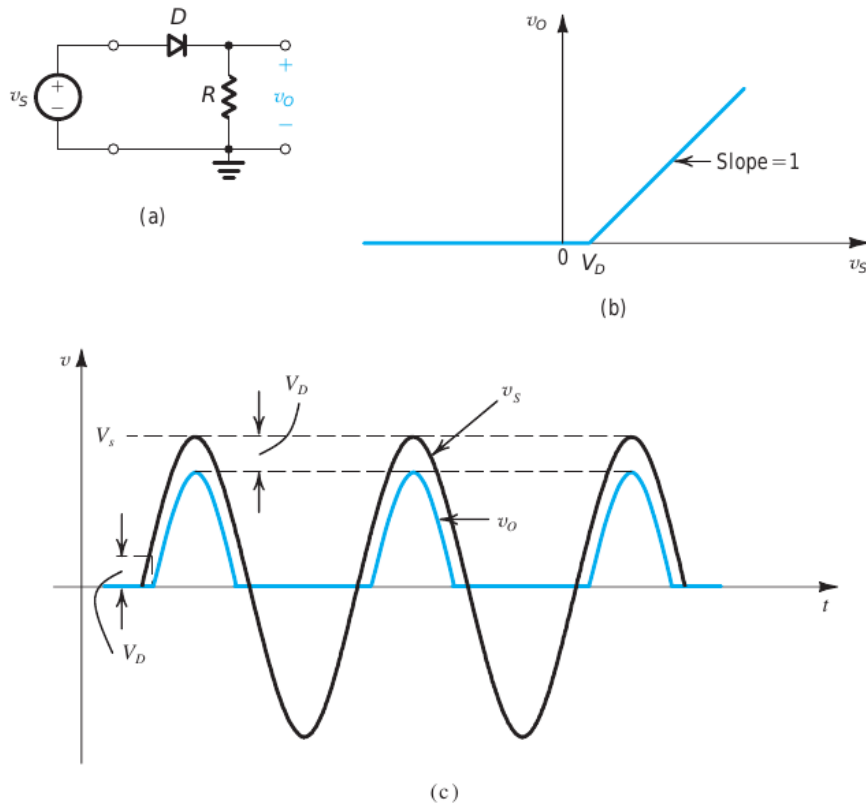
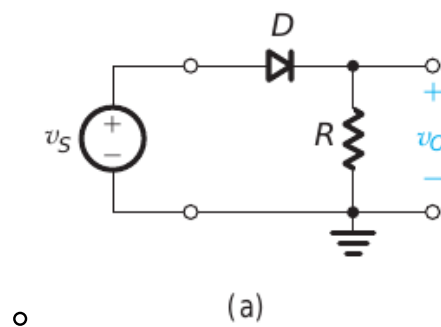
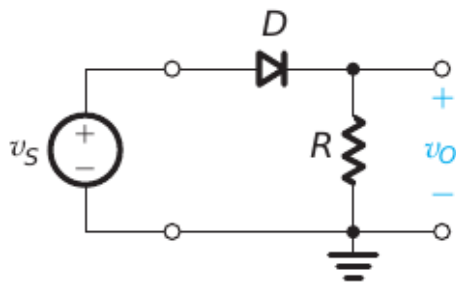


Figure 4.21 (a) Half-wave rectifier. (b) Transfer characteristic of the rectifier circuit. (c) Input and output

- waveforms.
- While selecting diodes for rectifier applications, two important parameters must be specified
 1. The current handling capability of the diode
 - which is the largest current the diode is expected to conduct
 2. The peak inverse voltage (PIV) that the diode must be able to withstand without breakdown
 - which is the largest reverse voltage that appears across the diode





(a)

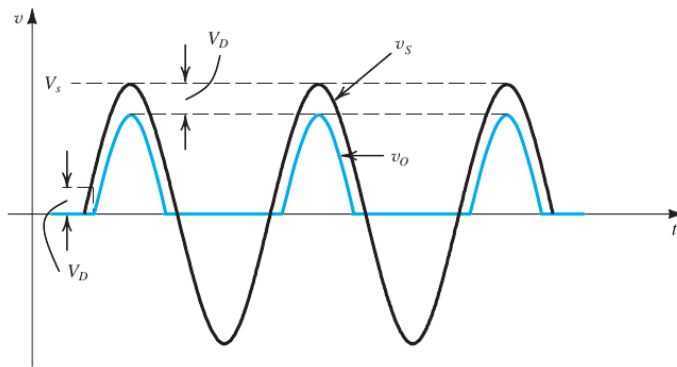
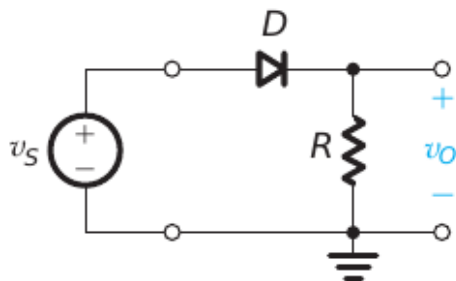


Figure 4.21 (c) Input and output waveforms. (c)

- Note that when v_S is -ve,
 - the diode will be cutoff
 - and v_O will be zero
- $\Rightarrow$ PIV equals the peak of v_S
- i.e. $PIV = V_S$

Exercise 4.19

- For the half-wave rectifier circuit in Fig 4.21(a), show the following: (a) For the half-cycles during which the diode conducts, conduction begins at an angle $\theta = \sin^{-1}(V_D/V_S)$ and terminates at $(\pi - \theta)$, for a total conduction angle of $(\pi - 2\theta)$. (b) The average value (dc component) of v_O is $V_O \approx (1/\pi)V_S - V_D/2$. (c) The peak diode current is $(V_S - V_D)/R$.
- Find numerical values for these quantities for the case of 12V (rms) sinusoidal input, $V_D = 0.7V$, and $R = 10\Omega$. Also give the value of PIV.

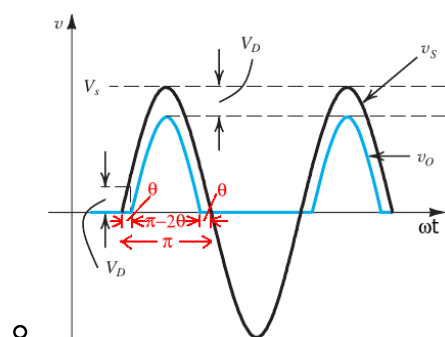


(a)

Exercise 4.19a

- For the half-wave rectifier circuit in Fig 4.21(a), show the following: (a) For the half-cycles during which the diode conducts, conduction begins at an angle

$\theta = \sin^{-1}(V_D/V_S)$ and terminates at $(\pi - \theta)$, for a total conduction angle of $(\pi - 2\theta)$.



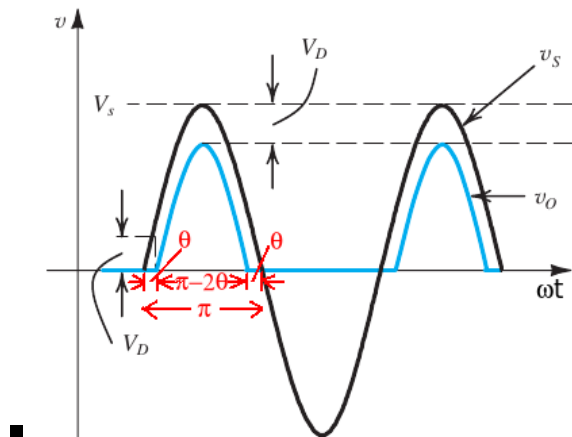
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o diode will conduct when $v_S \geq V_D$ o when $\omega t = \theta \Rightarrow v_S = V_D$

$$\blacksquare v_S = V_S \sin \theta = V_D \Rightarrow \theta = \sin^{-1}\left(\frac{V_D}{V_S}\right)$$

• diode will conduct when $v_S \geq V_D$ o when $\omega t = \theta \Rightarrow v_S = V_D$

$$\blacksquare v_S = V_S \sin \theta = V_D \Rightarrow \theta = \sin^{-1}\left(\frac{V_D}{V_S}\right)$$



■

o from the figure diode conducts for angle of $\pi - 2\theta$ o for $V_D = 0.7V$, $V_S = 12\sqrt{2} = 16.97V$

$$\blacksquare \theta = \sin^{-1}\left(\frac{V_D}{V_S}\right) = \sin^{-1}\left(\frac{0.7}{16.97}\right) = 0.041 \text{ rad} = 2.364^\circ$$

$$\blacksquare \Rightarrow \text{conduction angle} = \pi - 2\theta = 180^\circ - 2(2.364)^\circ = 175.272^\circ$$

Exercise 4.19b

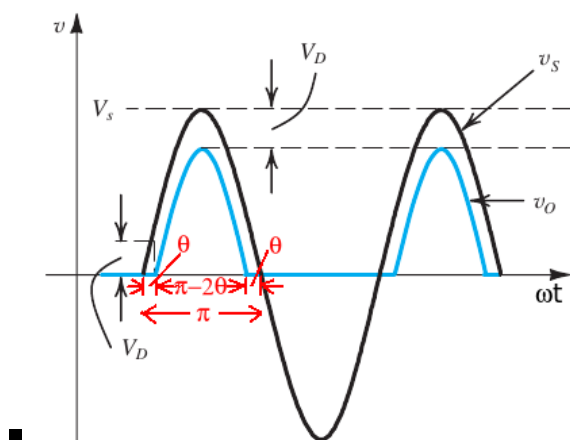
• For the half-wave rectifier circuit in Fig 4.21(a), show the following: (b) The average value (dc component) of v_O is $V_O \approx (1/\pi)V_S - V_D/2$.

o $v_{O,ave} = v_{O,average} = ?$

$$\blacksquare v_{O,ave} = \frac{1}{T} \int_0^T v_O(t) dt = \frac{1}{2\pi} \int_0^{2\pi} v_O(\omega t) d(\omega t)$$

$$\blacksquare \text{from fig } v_O(\omega t) = \begin{cases} 0 & , 0 \leq \omega t \leq \theta \\ v_S - V_D & , \theta \leq \omega t \leq \pi - \theta \\ 0 & , \pi - \theta \leq \omega t \leq 2\pi \end{cases}$$

$$\blacksquare v_S = V_S \sin(\omega t)$$



$$\circ \Rightarrow v_{O,ave} = \frac{1}{2\pi} \int_0^{2\pi} v_O(\omega t) d(\omega t) = \frac{1}{2\pi} \int_{\theta}^{\pi-\theta} \{ V_S \sin(\omega t) - V_D \} d(\omega t)$$

$$\bullet v_{O,ave} = \frac{1}{2\pi} \int_{\theta}^{\pi-\theta} \{ V_S \sin(\omega t) - V_D \} d(\omega t)$$

$$\circ v_{O,ave} = \frac{1}{2\pi} \left[\int_{\theta}^{\pi-\theta} V_S \sin(\omega t) d(\omega t) - \int_{\theta}^{\pi-\theta} V_D d(\omega t) \right]$$

$$\circ v_{O,ave} = \frac{1}{2\pi} \left[V_S [-\cos(\omega t)]_{\theta}^{\pi-\theta} - V_D (\pi - \theta - \theta) \right]$$

$$\circ v_{O,ave} = \frac{1}{2\pi} \left[V_S [-\cos(\pi - \theta) + \cos \theta] - V_D (\pi - 2\theta) \right]$$

$$\circ \text{ for } \theta \rightarrow 0 \Rightarrow \pi - 2\theta \approx \pi, \cos(\theta) \approx 1 \text{ and } \cos(\pi - \theta) \approx \cos(\pi) = -1$$

$$\blacksquare v_{O,ave} = \frac{1}{2\pi} [V_S [-(-1) + 1] - V_D (\pi)] = \frac{1}{2\pi} [2V_S - V_D (\pi)]$$

$$\blacksquare v_{O,ave} = \frac{V_S}{\pi} - \frac{V_D}{2}$$

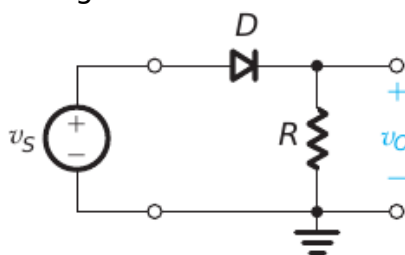
$$\circ \text{ for } V_S = 16.97V \text{ and } V_D = 0.7V$$

$$\blacksquare \Rightarrow v_{O,ave} = \frac{V_S}{\pi} - \frac{V_D}{2} = \frac{16.97}{\pi} - \frac{0.7}{2} = 5.0517V$$

Exercise 4.19c

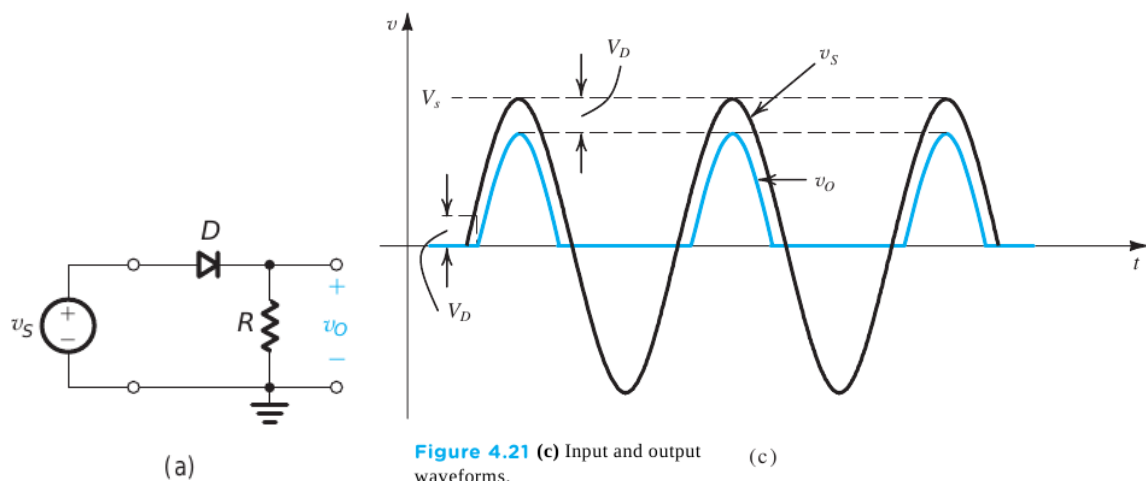
- For the half-wave rectifier circuit in Fig 4.21(a), show the following: (c) The peak diode current is $(V_S - V_D)/R$.

- Also give the value of PIV.



(a)

- peak diode current occurs when the diode is conducting
 - and the load is getting the peak current
 - i.e. when v_o is at maximum
- peak output voltage is $V_S - V_D$

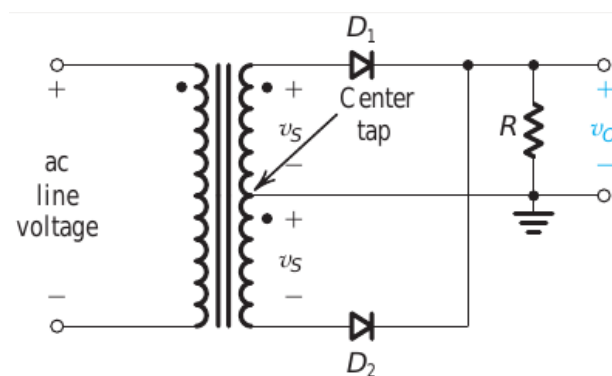


- peak output voltage is $V_s - V_D$
- $\Rightarrow$ peak output current = peak diode current

$$= \frac{V_s - V_D}{R} = \frac{16.97 - 0.7}{100} = 162.7 \text{ mA}$$
- $PIV = V_s = 16.97 \text{ V}$

The Full-Wave Rectifier using a Center-tapped Transformer

- utilizes both halves of the input sinusoid
 - to provide unipolar output, it inverts the negative halves of the input sine wave
 - One way to implement such a full wave rectifier is to
 - utilize a transformer with center-tapped secondary winding



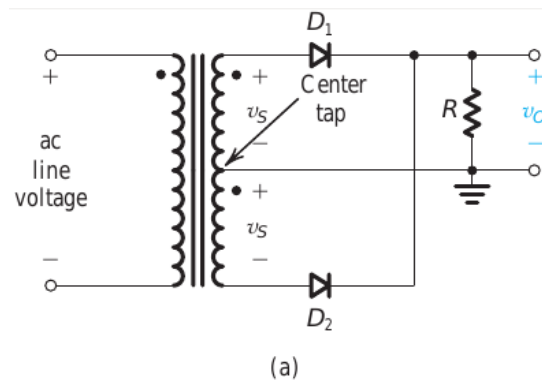
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(a)

- the transformer secondary winding is center-tapped to provide two equal voltages v_s

when the input line voltage is +ve

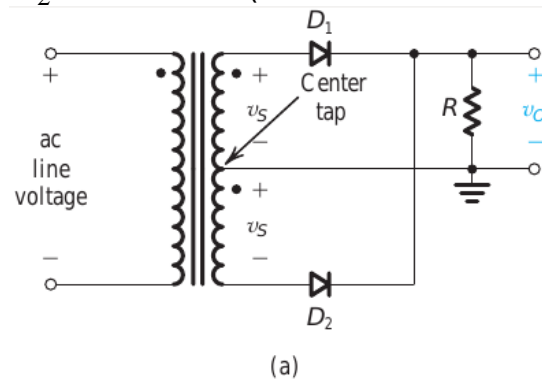
- both the voltages labeled v_s will be positive
 - $\Rightarrow D_1$ will conduct
 - and D_2 will be reverse biased.



- the current through D_1 will flow through R and then back to the center tap
 - thus the circuit behaves as a half-wave rectifier
 - and the diode D_1 conduct during the +ve half cycle of line voltage

when the input line voltage is -ve

- both the voltages labeled v_S will be negative
 - $\Rightarrow D_1$ will be reverse biased.
 - D_2 will conduct (forward biased)



- the current conducted through D_2 will flow through R and then back to the center tap
 - Thus the circuit is again acting as a half wave rectifier
 - during the negative-cycle with D_2 forward-biased
- Note that the current through R always flows in the same direction
 - $\Rightarrow v_O$ will be unipolar
 - Using the constant voltage drop model for the diodes,
 - the transfer characteristic of this rectifier is

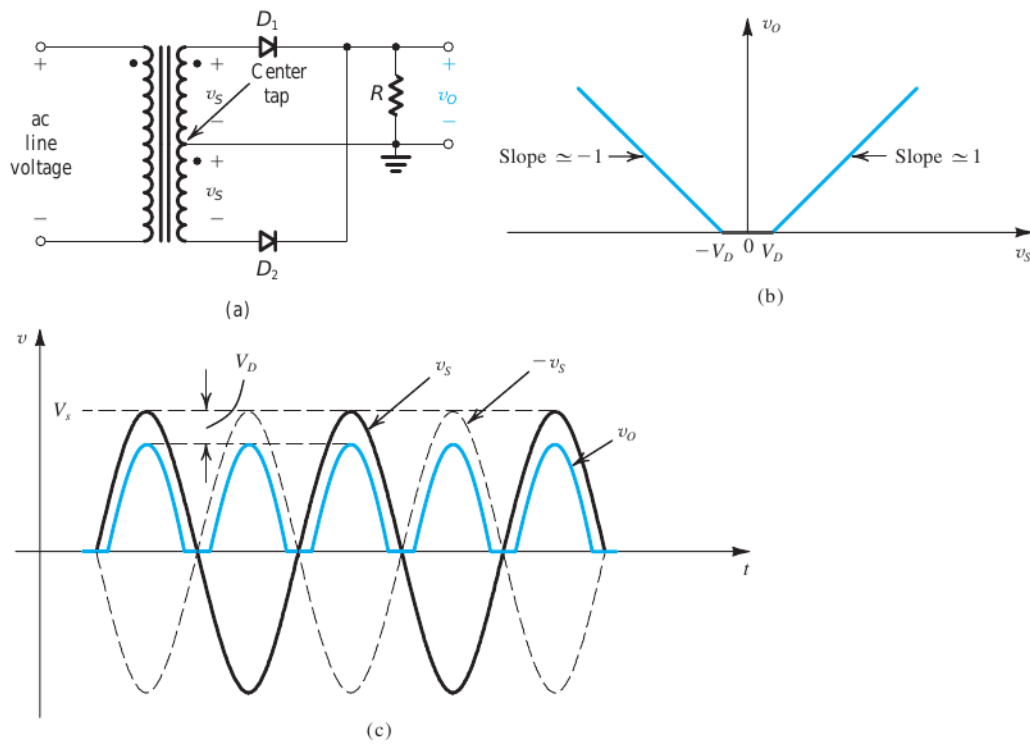


Figure 4.22 Full-wave rectifier utilizing a transformer with a center-tapped secondary winding: (a) circuit; (b) transfer characteristic assuming a constant-voltage-drop model for the diodes; (c) input and output waveforms.

PIV (Peak Inverse Voltage)

- during the +ve half cycle of line voltage,
 - D_1 is conducting and D_2 is reverse-biased (cut-off)

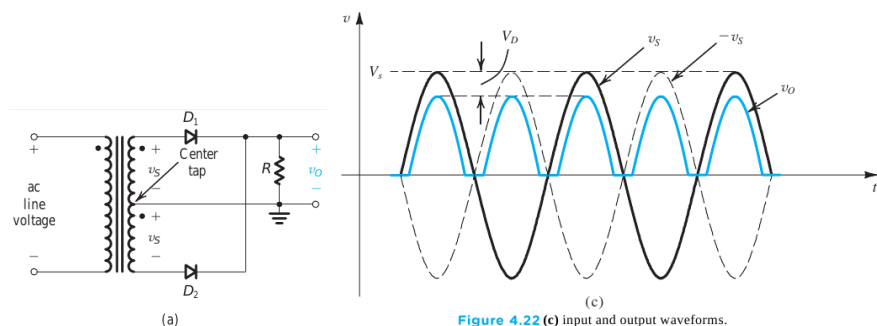


Figure 4.22 (c) input and output waveforms.

- PIV is the maximum reverse voltage across the diode D_2
 - the voltage at the cathode of D_2 is $v_O = v_S - V_D$ and $v_{O,max} = V_S - V_D$
 - the voltage at the anode of D_2 is $-v_S$ and when v_O is maximum $\Rightarrow -v_S$ is at $-V_S$
 - $\Rightarrow PIV = \text{maximum reverse voltage across } D_2$
 - $PIV = \text{max voltage at cathode} - \text{max voltage at anode}$
 - $PIV = (V_S - V_D) - (-V_S) = 2V_S - V_D$

The Bridge Rectifier

- A full wave rectifier can also be implemented using 4 diodes

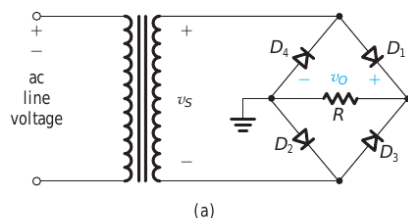


Figure 4.23 The bridge rectifier: (a) circuit;

- o this bridge rectifier does not require a center-tapped transformer
- o this rectifier however requires four diodes
 - instead of 2 in case of a full wave rectifier.
- o this is not much a disadvantage, as diodes are inexpensive and a diode bridge can be purchased in a single package
- o the operation of the bridge rectifier can be explained as follows

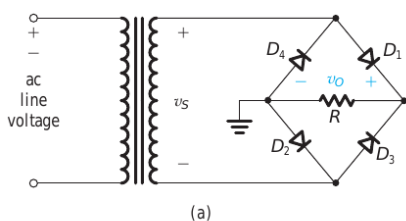
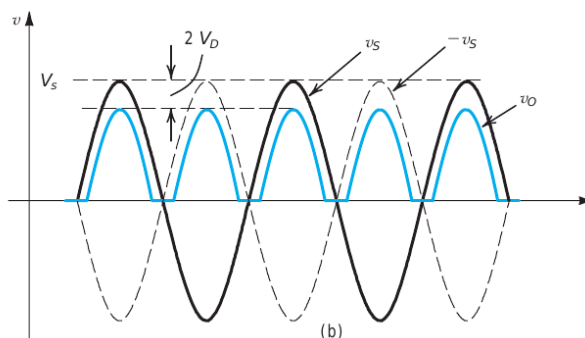
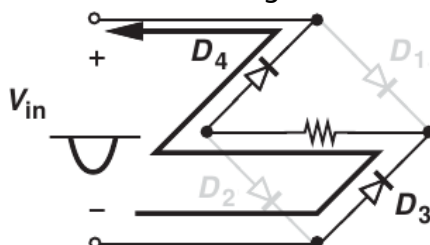


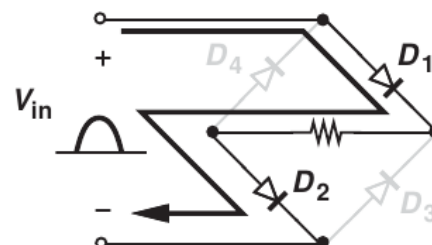
Figure 4.23 The bridge rectifier: (a) circuit; (b) input and output waveforms.



- During the positive half-cycle of the input-line voltage
 - o v_S is positive $\Rightarrow$ diodes D_1 and D_2 are forward-biased
 - D_3 and D_4 are reverse-biased
 - so the current flows through D_1 , resistor R and D_2 .
 - o Note that there are two diodes in series in the conduction path.
 - o $\Rightarrow v_O$ will be lower than v_S by two diode drops
 - $v_O = v_S - 2V_D$
- During the negative half-cycle of the input-line voltage
 - o v_S is negative and $-v_S$ is positive
 - $\Rightarrow$ diodes D_3 and D_4 are forward-biased
 - while D_1 and D_2 are reverse-biased
 - so the current flows through D_3 , resistor R and D_4 .
- Note that during both half cycles,
 - o current flows through R in the same direction



(c)



(d)

- o (c) current path for negative input,
- o $\Rightarrow v_O$ will always be positive

(d) current path for positive input.

- v_O is unipolar

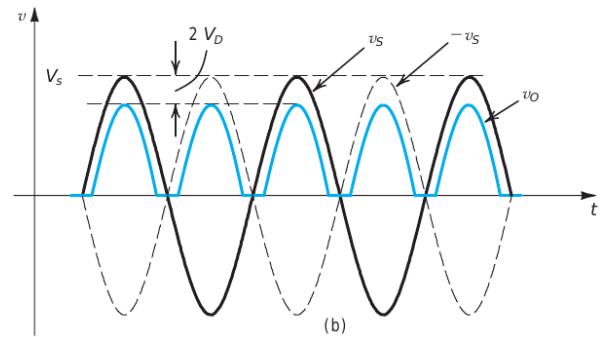
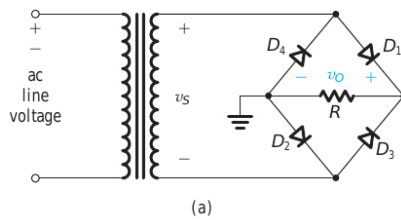


Figure 4.23 The bridge rectifier: (a) circuit; (b) input and output waveforms.

o

PIV (Peak Inverse Voltage)

- During the +ve half cycle of line voltage

- o D_1, D_2 are conducting
- o D_3, D_4 are reverse biased

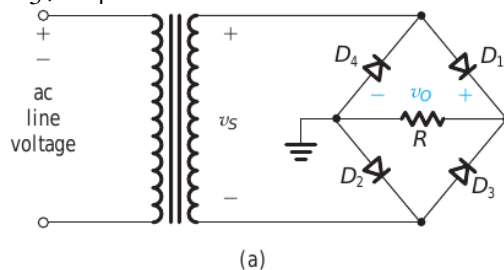


Figure 4.23 The bridge rectifier: (a) circuit;

- o for Diode D_3 , KVL along the loop formed by D_3 , R and D_2

$$\blacksquare \Rightarrow V_{D2,on} + v_O = v_{D3,off}$$

$$\blacksquare \Rightarrow V_D + v_O = v_{D3,off} \because V_{D2,on} = V_D$$

- o $v_{D3,off} = V_D + v_O$

- o the maximum value of $v_{D3,off}$ occurs when v_O is at its peak value of $V_S - 2V_D$

$$\blacksquare \Rightarrow PIV \text{ for } D_3 = V_D + (V_S - 2V_D) = V_S - V_D$$

- o Note that PIV for other diodes will be the same as for D_3

- PIV for bridge rectifier = $V_S - V_D$

- o PIV for full wave rectifier with center-tapped transformer = $2V_S - V_D$

- o $\Rightarrow \{PIV \text{ for bridge rectifier}\} \approx \text{half of } \{PIV \text{ for full wave rectifier with center-tapped transformer}\}$